

Böl Yönet yaklaşımı ile sıralama(Örnek:Merge Sort)

- Sıralanması istenen kümeler önce iki alt kümeye ayrılır.
- Fonksiyon kendini sol ve sağ alt küme için iki kez çağırır.
- Parçalama işlemine altkümedeki eleman sayısı bir oluncaya kadar devam edilir.
- Daha sonra rekürsif çağrılar geriye doğru çağırana dönmeye başlar ve bu sırada altkümeler elemanları sıralı olacak biçimde birleştirilir.

Birleştirmeli Sırlama (Merge Sort)

- Böl-Yönet yaklaşımli sıralama
- Her bir altdizide bir eleman kalıncaya kadar sırasız diziyi ikiye böl
- Alt problem çözümlerini birleştir:
 - Altdizilerin ilk elemanlarını karşılaştır.
 - En küçük elemanı kaldır ve sonuç diziye koy
 - Bütün elemanlar sonuç diziye konuncaya kadar işlemleri tekrarla

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Birleştirme (Merge Sort) Algoritması

MERGE_SORT(A, p, r)

if $p < r$

then $q \leftarrow \lfloor (p+r) / 2 \rfloor$

MERGE_SORT(A, p, q)

MERGE_SORT(A, q+1, r)

MERGE(A, p, q, r)

Merge Algorithm

MERGE(A, p, q, r)

$n_1 \leftarrow q - p + 1$

$n_2 \leftarrow r - q$

// Create arrays L[1.. n_1+1] and R[1.. n_2+1]

for i $\leftarrow$ 1 **to** n_1

do L[i] $\leftarrow$ A[p + i - 1]

for j $\leftarrow$ 1 **to** n_2

do R[j] $\leftarrow$ A[q + j]

L[$n_1 + 1$] $\leftarrow \infty$; R[$n_2 + 1$] $\leftarrow \infty$

i $\leftarrow$ 1; j $\leftarrow$ 1

for k $\leftarrow$ p **to** r

do if L[i] $\leq$ R[j]

then A[k] $\leftarrow$ L[i]

 i $\leftarrow$ i + 1

else A[k] $\leftarrow$ R[j]

 j $\leftarrow$ j + 1

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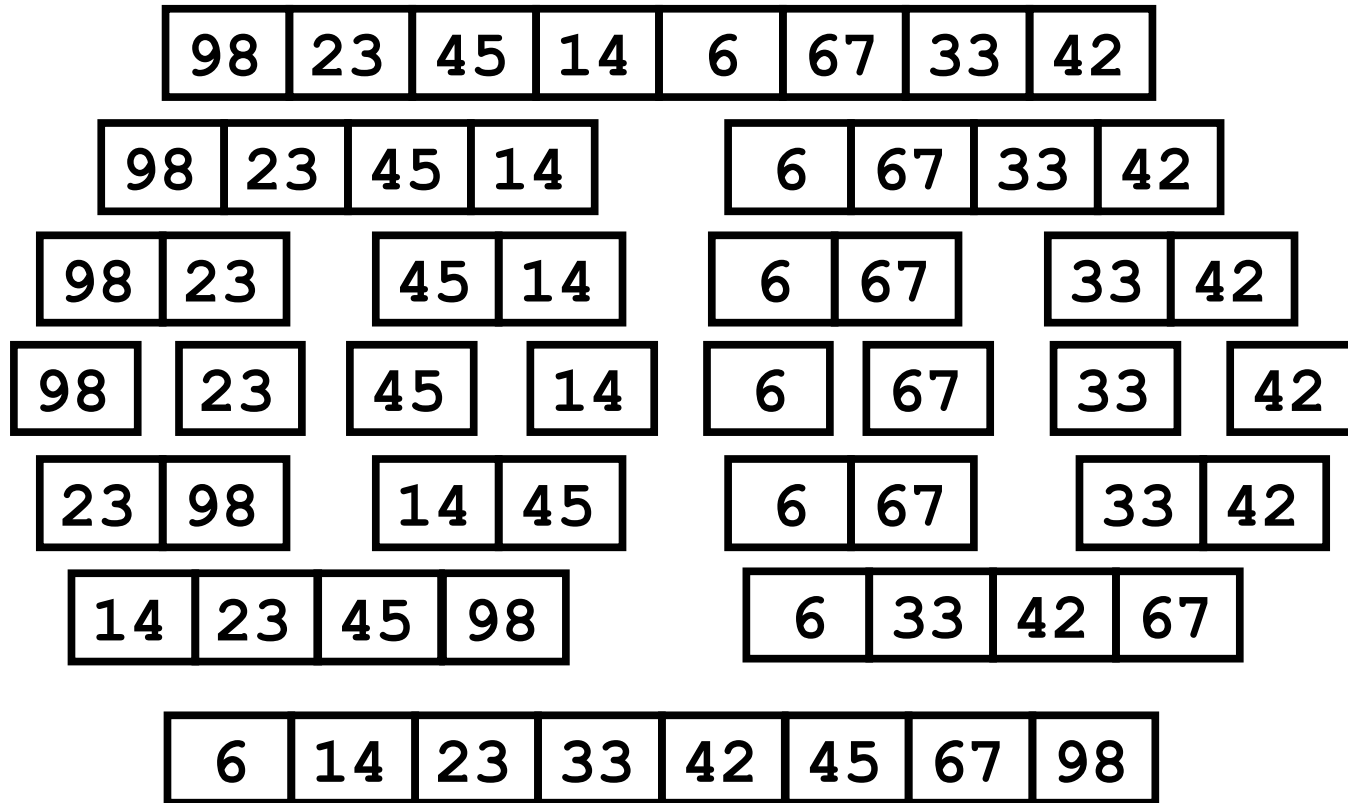
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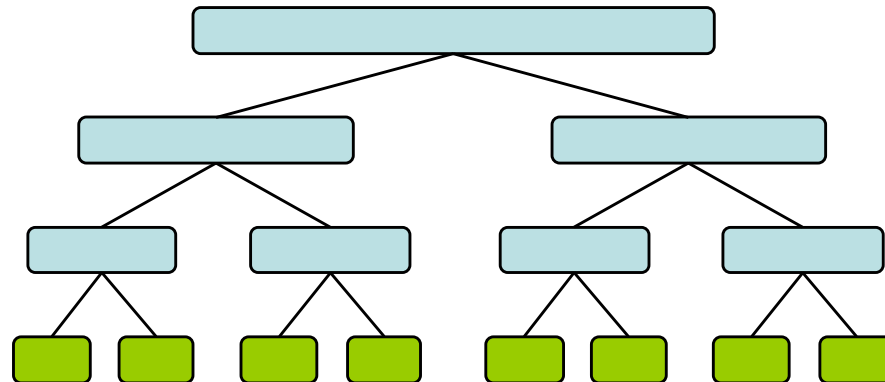
Analiz

- Her rekürsif çağrıda diziyi iki parçaya bölüyoruz. Oluşan ağacın yüksekliği $O(\log n)$ olacaktır.
- i derinlikli düğümde yapılan işlemler $O(n)$ dir.
 - $n/2^i$ elemanlı 2^i parçalama ve birleştirme yaparız.
 - 2^{i+1} rekürsif çağrı yapılır.
- Toplam $O(n \log n)$ işlem yaparız.

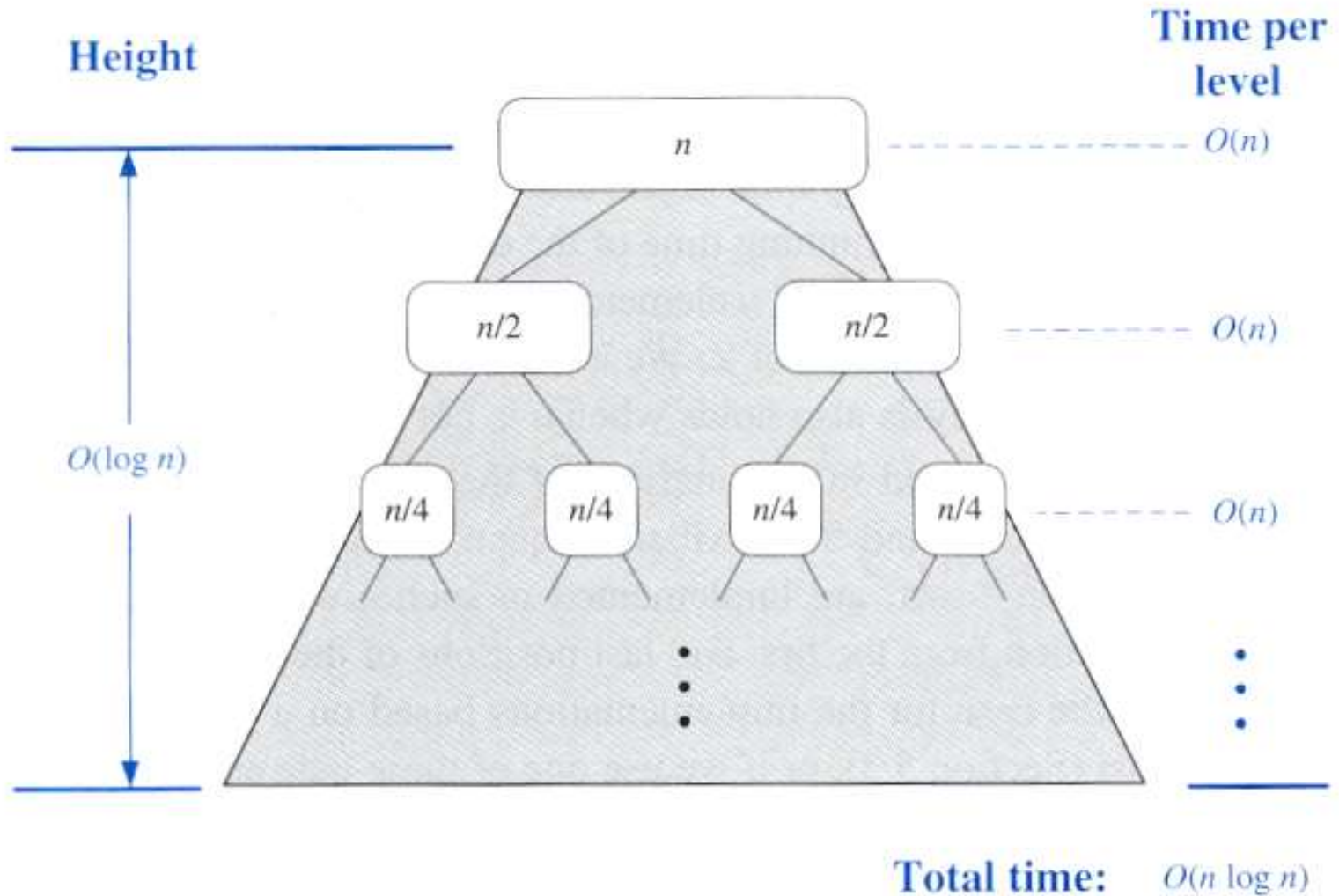
depth #seqs size

$$0 \qquad 1 \qquad n$$
$$1 \qquad 2 \qquad n/2$$
$$i \quad 2^i \quad n/2^i$$

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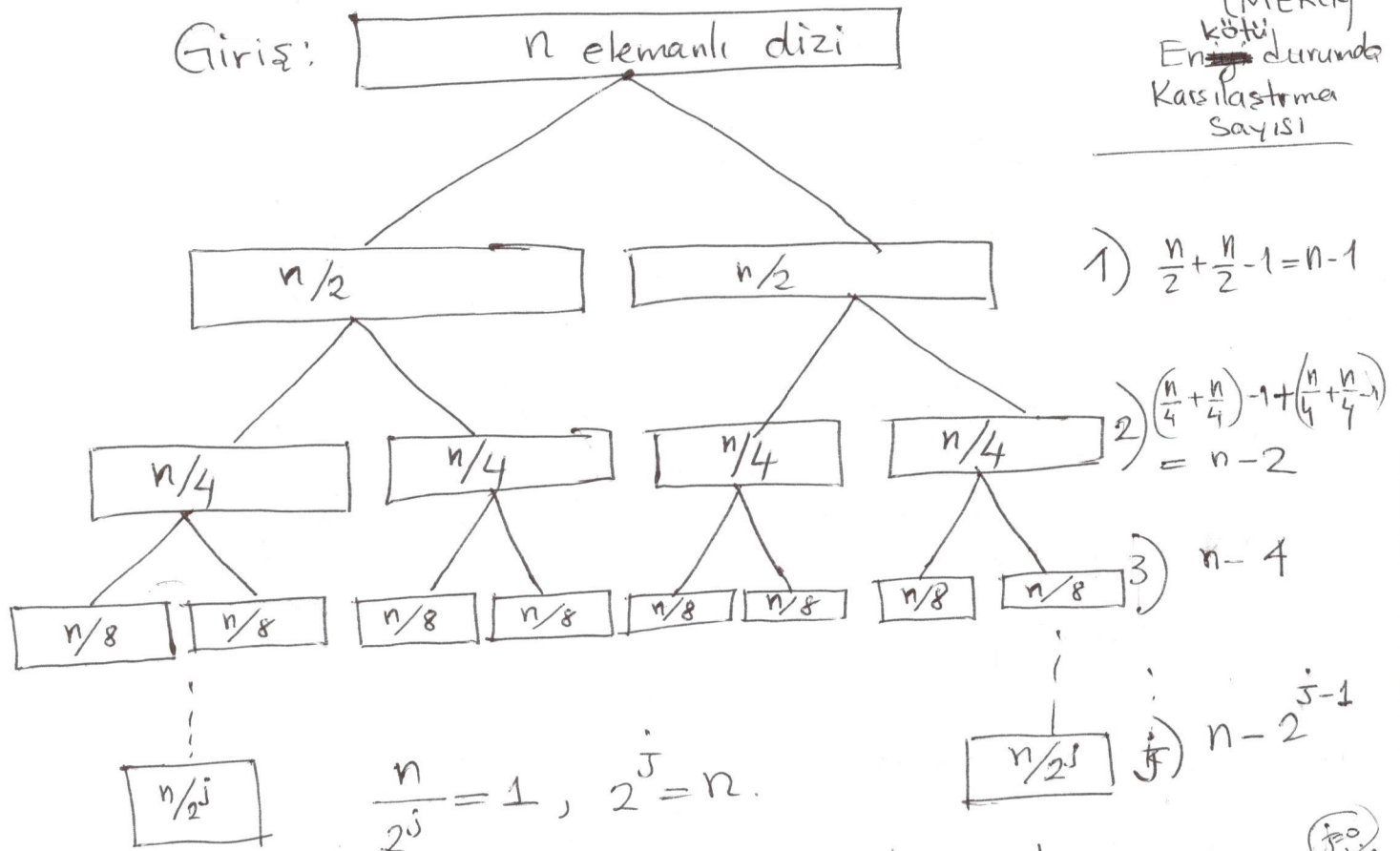


Analiz



(Topdown) Merge Sort Algorithm

(MERGE)
En ~~kötü~~ durumda
Karşılaştım
Sayısı



$$\sum_{j=1}^k (n - 2^{j-1}) = \sum_{j=1}^k n - \sum_{j=1}^k 2^{j-1} = \sum_{j=1}^k n - \frac{1}{2} \sum_{j=1}^k 2^j = kn - \frac{1}{2} [2^{k+1} - 1 - 1]$$

$$= kn - \frac{1}{2} [2^{k+1} - 2]$$

$$= kn - \frac{1}{2} [2(2^k - 1)]$$

$$= kn - 2^k + 1$$

$$= n \cdot \log_2 n - n + 1 \checkmark$$

MERGE işlemindeki maksimum karşılaştırma sayısı $n \cdot n$

$$n_1 + n_2 - 1 = \frac{n}{2} + \frac{n}{2} - 1 = n - 1$$

minimum

Karsilastırma sayın

n_1 idi

Burada $\frac{n}{2}$ olur.

Hatırlatma:

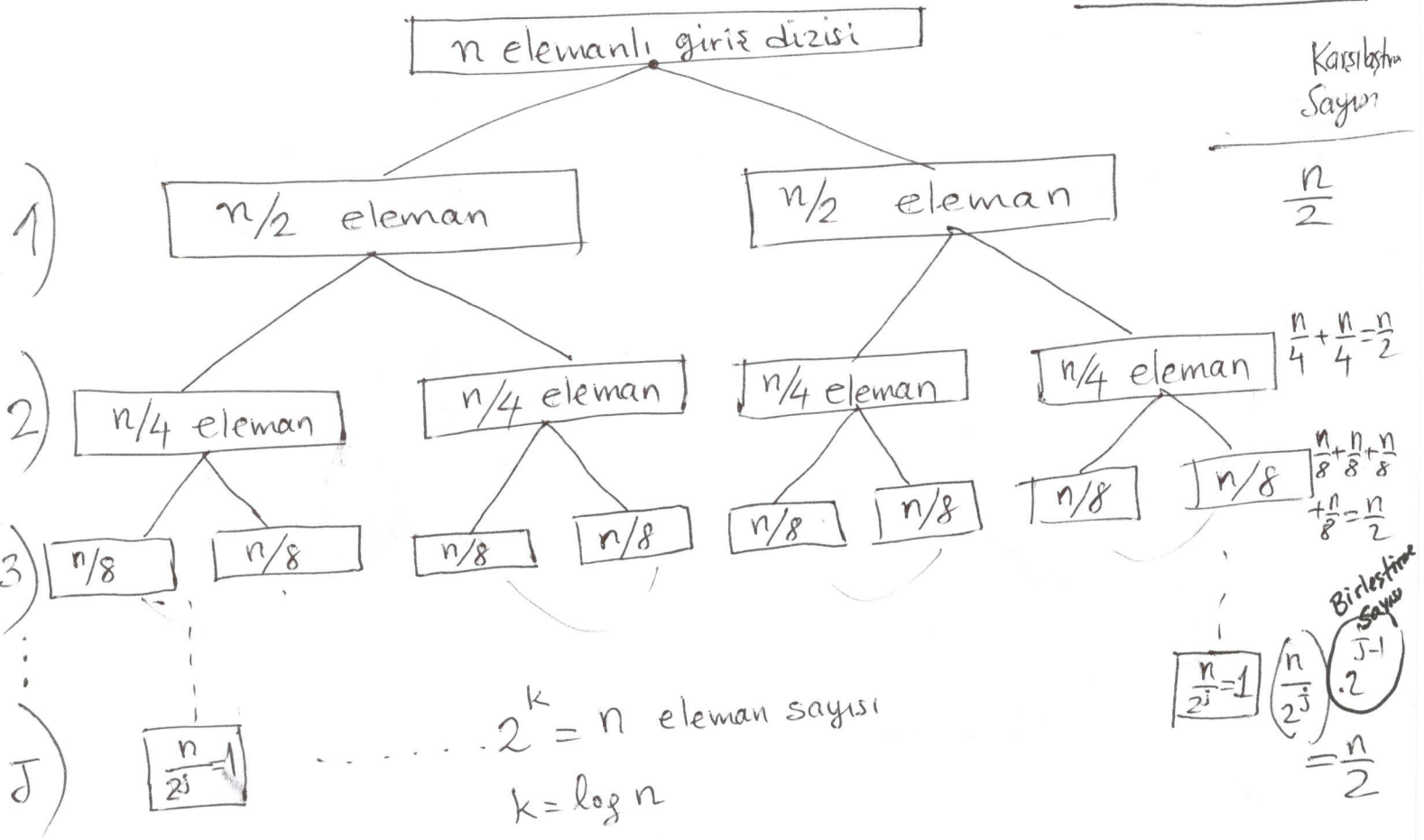
$$\sum_{j=0}^n a^j = \frac{a^{n+1} - 1}{a - 1}$$

$$\sum_{j=1}^n a^j = \frac{a^{n+1} - 1}{a - 1} - 1$$

Δa^0
Çıkarıldı.

(1)

(Topdown) Merge Sort Algoritması (analitik) ^{iyi} ~~en kötü~~ durum



$$\sum_{j=1}^k \left(\frac{n}{2} \right) = \frac{kn}{2} = \frac{n \cdot \log n}{2}$$

Açıklama

1. adımda	$\frac{n}{2}$ eleman	2 dizi içi	1 MERGE	1. $\frac{n}{2} = \frac{n}{2}$
2. adımda	$\frac{n}{4}$ eleman	4 dizi içi	2 MERGE	2. $\frac{n}{4} = \frac{n}{2}$
3. adımda	$\frac{n}{8}$ eleman	8 dizi içi	4 MERGE	4. $\frac{n}{8} = \frac{n}{2}$
J . adımda	$\frac{n}{2^j}$ eleman	2^j dizi içi	2^{j-1} MERGE	$2^{j-1} \cdot \frac{n}{2^j} = \frac{n}{2}$

$$+ \sum_{j=1}^k \left(\frac{n}{2} \right) = \frac{kn}{2} = \frac{n \log n}{2}$$

1. iterasyonda 1 elemanlı n adet sıralı dizi birleştiriliyor.
2. iterasyonda 2 elemanlı $\frac{n}{2}$ adet sıralı dizi birleştiriliyor
 $\frac{n}{4}$ merge (2 ile 3) arasında karşılaştırmalar.
3. iterasyonda 4 elemanlı $\frac{n}{4}$ adet sıralı dizi birleştiriliyor.
 $\frac{n}{8}$ merge (4 ile 7) arasında karşılaştırmalar.
- ...
- j . iterasyonda 2^{j-1} elemanlı $\frac{n}{2^{j-1}}$ adet sıralı dizi birleştiriliyor.
 $\frac{n}{2^j}$ merge 2^{j-1} ile $2 \cdot 2^{j-1} - 1$

j . adımdaki karşılaştırma sayısı;

$$\frac{n}{2^j} \cdot \left(2^{j-1} \right) \text{ ile } \frac{n}{2^j} \left(2^{j-1} \cdot 2 - 1 \right) = \frac{n}{2^j} (2^j - 1) = n - \frac{n}{2^j} \text{ arasında dır.}$$

Algoritmadaki dış döngü $k = \log n$ defa çalışacağından, karşılaştırma sayısı

$$\sum_{j=1}^k \frac{n}{2^j} \cdot 2^{j-1} = \sum_{j=1}^k \frac{n}{2} = \frac{kn}{2} = \frac{n \log n}{2} \text{ olur. minimum karşılaştırma sayısı}$$

maksimum karşılaştırma sayısı ise;

$$\sum_{j=1}^k n - \frac{n}{2^j} = \sum_{j=1}^k n - \sum_{j=1}^k \frac{n}{2^j} = kn - n \left(\sum_{j=1}^k \left(\frac{1}{2} \right)^j \right) = kn - n \cdot \left(\frac{n-1}{n} \right) = n \log n - n + 1$$

$$\sum_{i=0}^n a^i = 1 + a + a^2 + \dots + a^n = \frac{a^{n+1} - 1}{a - 1}, a \neq 1 \text{ olduğundan bulunur.}$$

$$\sum_{j=1}^k \left(\frac{1}{2} \right)^j = \frac{\left(\frac{1}{2} \right)^{k+1} - 1}{\frac{1}{2} - 1} = -2 \left(\frac{1}{2^{k+1}} - 1 \right) - 1 = -2 + 1 = 1 - \frac{1}{2^k} = 1 - \frac{1}{2^{\log n}} = 1 - \frac{1}{n} = \frac{n-1}{n}$$

en son $\frac{2^{j-1}}{2} = \frac{n}{2}$ yada $\frac{n}{2^{j-1}} = 2$ olunca isten durur

$$2^j = n, j = \log_2 n$$

$$j = \log_2 n$$

(5) daha düzenli